

Evolutionary Optimization of Physician Scheduling via Evolutionary Computation with Bipartite Matching and Machine Learning

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1 Introduction

Physician scheduling is a critical component of healthcare operations, requiring the assignment of physicians to clinic sessions while ensuring adequate patient access and service coverage within the clinic setting. The scheduling process must satisfy a range of operational constraints, including physician availability, specialty requirements, workload limits, and institutional policies. At the same time, it must balance competing objectives such as fairness, recovery time, and workload sustainability.

Poorly structured schedules can lead to excessive work hours, insufficient recovery time, and increased fatigue among physicians. These factors have been associated with burnout and increased mental health risks among physicians, highlighting the importance of designing scheduling systems that support sustainable workloads and adequate recovery. These issues can negatively affect physician well-being and may impact the quality and safety of patient care.

Traditional scheduling approaches are typically based on manual planning or fixed heuristic rules. While these methods can produce feasible schedules, they often fail to systematically explore trade-offs between competing objectives or adapt to changing conditions. As healthcare systems become more complex, there is a growing need for computational methods that can improve scheduling quality while maintaining interpretability and feasibility.

This research proposes a hybrid optimization framework that combines bipartite matching, evolutionary computation, and machine learning to improve physician scheduling. The approach aims to generate schedules that balance coverage requirements with fairness, recovery, and fatigue-related considerations.

2 Problem Statement

The physician scheduling problem involves assigning a set of physicians to a set of clinic sessions over a planning horizon.

This assignment must satisfy the following:

Hard constraints (feasibility requirements):

- physician availability
- qualification and specialty requirements
- maximum working hours per week
- minimum rest periods between clinic sessions
- maximum number of consecutive workdays
- required clinic session coverage (each session must be fully staffed)

Soft constraints (optimization objectives):

- workload balance across physicians
- sufficient recovery time between assigned clinic sessions
- limitation of consecutive clinic days
- physician clinic session preferences

The problem is inherently multi-objective and combinatorial, as improving one objective may negatively impact another. Therefore, the goal is to design a system that ensures feasibility while optimizing these competing objectives.

3 Related Work

Physician scheduling has been widely studied within the field of workforce rostering and healthcare operations. Traditional approaches rely on heuristic and rule-based methods designed to ensure feasibility and coverage. While effective in practice, these approaches often fail to optimize competing objectives such as fairness and workload balance.

Optimization-based methods, including integer programming and constraint programming, have been applied to scheduling problems. These approaches are effective for enforcing hard constraints but can become computationally expensive and less flexible when incorporating multiple soft objectives.

Evolutionary algorithms have been proposed as an alternative for solving complex scheduling problems. Prior work has demonstrated that evolutionary computation can effectively explore large solution spaces and optimize multiple objectives simultaneously, particularly in improving workload balance and fairness in physician scheduling.

However, many existing approaches optimize schedules directly, which can reduce interpretability and complicate constraint enforcement. Matching-based approaches provide a structured alternative by modeling scheduling as an assignment problem between physicians and clinic sessions. Despite their advantages, these approaches typically rely on fixed cost structures and lack adaptive optimization capabilities.

Additionally, research has emphasized the importance of incorporating fatigue and recovery considerations. Studies have shown that limiting consecutive work periods and ensuring adequate rest are essential for maintaining physician performance and well-being. However, many models do not explicitly capture temporal workload patterns.

Recent work has also explored the integration of machine learning into optimization frameworks, particularly through surrogate models that approximate objective functions and reduce computational cost.

4 Research Gap

Despite existing work in physician scheduling, several limitations remain:

- insufficient integration of fatigue and recovery modeling
- reliance on static scheduling policies
- limited exploration of multi-objective trade-offs
- lack of interpretable and adaptive frameworks

This research addresses these gaps by proposing a framework that:

- explicitly models fatigue and recovery
- uses evolutionary computation to optimize scheduling policies
- maintains interpretability through matching-based construction
- incorporates machine learning to improve scalability

5 Research Questions

RQ1: Can evolutionary computation improve physician scheduling policies by optimizing interpretable cost structures used in matching-based scheduling?

RQ2: How can fatigue and recovery considerations be effectively incorporated into scheduling models?

RQ3: What trade-offs exist between coverage, fairness, and recovery, and how can they be explored using multi-objective optimization?

RQ4: Can machine learning surrogate models reduce the computational cost of evaluating scheduling policies?

RQ5 (Extension): How does incorporating physician–patient compatibility affect scheduling outcomes?

6 Proposed Methodology

6.1 Matching-Based Schedule Construction

The scheduling problem will be modeled as a minimum-cost bipartite matching problem between physicians and clinic sessions.

- nodes represent physicians and clinic sessions
- edges represent feasible assignments

An edge is created only if all hard constraints are satisfied, including availability, specialty requirements, workload limits, and recovery constraints. This ensures that all generated schedules are feasible.

This formulation separates feasibility from optimization, allowing the model to maintain valid schedules while exploring trade-offs between competing objectives.

6.2 Cost Function Design

Each edge in the matching graph is assigned a cost representing violations of soft constraints:

- fatigue penalties based on recent workload
- recovery penalties for insufficient rest or fragmented time off
- fairness penalties to balance workload across physicians
- preference penalties for undesirable assignments

6.3 Evolutionary Optimization of Policies

Instead of optimizing schedules directly, the framework optimizes policy parameters.

Each candidate solution encodes weights controlling:

- fatigue importance
- recovery importance
- fairness penalties
- preference weighting

This research uses Evolutionary Computation as its main optimization framework, specifically a GA-based multi-objective evolutionary algorithm. The primary method will be NSGA-II, which is well suited for optimizing trade-offs between competing scheduling objectives such as coverage, fairness, recovery, and fatigue.

6.4 Fatigue and Recovery Modeling

Fatigue and recovery will be modeled using:

Hard constraints:

- maximum consecutive workdays
- minimum rest periods

Soft penalties:

- insufficient recovery time
- clustering of high-load or long sessions
- uneven workload distribution

6.5 Machine Learning Integration (Optional)

Machine learning will be used as a surrogate model to approximate scheduling performance and reduce computational cost during optimization.

6.6 Extension: Patient Integration

An optional extension will incorporate physician–patient compatibility, introducing a tri-partite assignment model.

7 Experimental Design

Baselines:

- heuristic scheduling methods
- fixed-weight matching

Metrics:

- coverage violations
- workload fairness
- recovery quality
- fatigue-risk proxies
- preference satisfaction

Experiments:

- varying demand scenarios
- physician availability changes
- robustness testing

8 Expected Contributions

- fatigue-aware physician scheduling framework
- evolutionary optimization of scheduling policies
- interpretable matching-based scheduling model
- empirical analysis of trade-offs between scheduling objectives
- extensible framework for future healthcare optimization

9 Feasibility

The project is feasible due to its modular structure. Core components (matching, evolutionary computation, and fatigue modeling) can be completed independently of optional extensions such as machine learning and patient integration.

10 Conclusion

This research proposes a structured approach to improving physician scheduling using optimization techniques. By combining bipartite matching with evolutionary computation, the framework aims to produce schedules that balance operational requirements with physician well-being. The proposed approach contributes to both theoretical research in optimization and practical applications in healthcare systems.